■ Mushroom Business Management & E-Commerce Platform (Mushroom ERP)

Document 03: Relational Database Schema & Comprehensive Data Dictionary

Project Name: SporeERP
Database Engine: MariaDB 11.x (InnoDB Engine)
Character Set: `utf8mb4` | **Collation:** `utf8mb4_unicode_ci`
ORM: Prisma ORM 5.x / 6.x
Document Code: DOC-03-DB-SCHEMA

1. High-Level Entity-Relationship (ER) Overview

The relational structure guarantees zero data anomalies and maintains referential integrity across the entire agronomy, supply chain, and e-commerce lifecycle:



2. Complete Data Dictionary by Module

2.1 Authentication & RBAC Module

Table: `users`

Stores all authentication principals (Staff, Founders, and Customers).

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique user identifier.
`name`	VARCHAR(100)	NOT NULL	None	Full legal name.
`email`	VARCHAR(150)	UNIQUE, NOT NULL	None	Login email address.
`phone`	VARCHAR(20)	UNIQUE, NULLABLE	NULL	Mobile phone number.
`password_hash`	VARCHAR(255)	NOT NULL	None	Bcrypt hashed password.
`is_active`	BOOLEAN	NOT NULL	TRUE	Account active state.
`email_verified`	DATETIME(3)	NULLABLE	NULL	Verification timestamp.

Column Name	Data Type	Constraints	Default Value	Description
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Record creation timestamp.
`updated_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Last modification timestamp.

Indexes: `idx_users_email` ON (`email`), `idx_users_phone` ON (`phone`).

Table: `roles`

Defines discrete system roles.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique role ID.
`name`	VARCHAR(50)	UNIQUE, NOT NULL	None	Role code (`FOUNDER_ADMIN` , `FARM_MANAGER` , etc.).
`display_name`	VARCHAR(100)	NOT NULL	None	Human-readable role title.
`description`	TEXT	NULLABLE	NULL	Detailed scope of responsibility.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Creation timestamp.

Table: `permissions`

Defines atomic resource permissions.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique permission ID.
`code`	VARCHAR(80)	UNIQUE, NOT NULL	None	e.g. `production:batch:create` , `finance:pnl:view` .
`module`	VARCHAR(50)	NOT NULL	None	Module group (`PRODUCTION` , `FINANCE` , etc.).
`description`	VARCHAR(255)	NOT NULL	None	Description of the permission.

Table: `role_permissions`

Maps permissions to roles.

Column Name	Data Type	Constraints	Description
`role_id`	VARCHAR(36)	FK -> `roles(id)` ON DELETE CASCADE	Target role ID.
`permission_id`	VARCHAR(36)	FK -> `permissions(id)` ON DELETE CASCADE	Target permission ID.

Primary Key: (`role_id` , `permission_id`).

Table: `user_roles`

Assigns roles to users.

Column Name	Data Type	Constraints	Description
`user_id`	VARCHAR(36)	FK -> `users(id)` ON DELETE CASCADE	Target user ID.
`role_id`	VARCHAR(36)	FK -> `roles(id)` ON DELETE CASCADE	Target role ID.

Primary Key: (`user_id` , `role_id`).

2.2 Customer & Address Module

Table: `customers`

Customer profile details, loyalty points, and B2B GST credentials.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique customer ID.
`user_id`	VARCHAR(36)	UNIQUE, FK -> `users(id)`	None	Associated user account.
`customer_type`	ENUM('RETAIL', 'B2B_RESTAURANT', 'WHOLESALE')	NOT NULL	'RETAIL'	Customer classification.
`company_name`	VARCHAR(150)	NULLABLE	NULL	Restaurant / Business name for B2B.
`gstin`	VARCHAR(15)	NULLABLE	NULL	GST Identification Number.
`credit_limit`	DECIMAL(10,2)	NOT NULL	0.00	Approved credit limit for B2B.
`total_orders`	INT UNSIGNED	NOT NULL	0	Lifetime delivered orders.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Registration timestamp.

Table: `customer_addresses`

Multiple shipping/billing addresses for customers.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique address ID.
`customer_id`	VARCHAR(36)	FK -> `customers(id)` ON DELETE CASCADE	None	Owning customer.
`address_type`	ENUM('HOME', 'WORK', 'RESTAURANT_KITCHEN' 'WAREHOUSE')	NOT NULL	'HOME'	Address classification.
`recipient_name`	VARCHAR(100)	NOT NULL	None	Name of person receiving parcel.
`recipient_phone`	VARCHAR(20)	NOT NULL	None	Contact number for delivery.

Column Name	Data Type	Constraints	Default Value	Description
`address_line1`	VARCHAR(255)	NOT NULL	None	House/Flat/Shop number, Building.
`address_line2`	VARCHAR(255)	NULLABLE	NULL	Street, Landmark, Area.
`city`	VARCHAR(80)	NOT NULL	None	City / Town.
`state`	VARCHAR(80)	NOT NULL	None	State / Province.
`pincode`	VARCHAR(10)	NOT NULL	None	Postal PIN code.
`latitude`	DECIMAL(10,8)	NULLABLE	NULL	GPS latitude for routing.
`longitude`	DECIMAL(11,8)	NULLABLE	NULL	GPS longitude for routing.
`is_default`	BOOLEAN	NOT NULL	FALSE	Default delivery address flag.

Indexes: `idx_cust_addr_pincode` ON (`pincode`), `idx_cust_addr_cust` ON (`customer_id`).

2.3 Product Catalog & Categories Module

Table: `categories`

Product taxonomy tree.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique category ID.
`name`	VARCHAR(100)	NOT NULL	None	e.g. "Fresh Mushrooms", "Mushroom Spawn".
`slug`	VARCHAR(120)	UNIQUE, NOT NULL	None	URL slug (`fresh-oyster-mushrooms`).
`description`	TEXT	NULLABLE	NULL	Category description for SEO.
`image_url`	VARCHAR(255)	NULLABLE	NULL	Header thumbnail image.
`parent_id`	VARCHAR(36)	NULLABLE, FK -> `categories(id)`	NULL	Self-referencing subcategory parent.
`is_active`	BOOLEAN	NOT NULL	TRUE	Active status flag.

Table: `products`

The core e-commerce product catalog.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique product ID.
`category_id`	VARCHAR(36)	FK -> `categories(id)` ON DELETE RESTRICT	None	Product category.
`name`	VARCHAR(150)	NOT NULL	None	e.g. "Fresh Blue Oyster Mushrooms".
`slug`	VARCHAR(180)	UNIQUE, NOT NULL	None	SEO URL slug.

Column Name	Data Type	Constraints	Default Value	Description
`sku`	VARCHAR(50)	UNIQUE, NOT NULL	None	Stock Keeping Unit code.
`description`	LONGTEXT	NOT NULL	None	Detailed rich-text description.
`unit_type`	ENUM('KG', 'GRAMS_200', 'GRAMS_500', 'PACK', 'BOTTLE')	NOT NULL	'KG'	Unit of measurement.
`retail_price`	DECIMAL(10,2)	NOT NULL	None	Standard retail selling price (Rs.).
`b2b_price`	DECIMAL(10,2)	NULLABLE	NULL	Discounted bulk price for restaurants.
`min_order_qty`	DECIMAL(8,2)	NOT NULL	1.00	Minimum order quantity.
`shelf_life_days`	INT UNSIGNED	NOT NULL	2	Perishability warning threshold.
`status`	ENUM('DRAFT', 'PUBLISHED', 'OUT_OF_STOCK', 'ARCHIVED')	NOT NULL	'DRAFT'	Publishing state.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Creation timestamp.
`updated_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Update timestamp.

Indexes: `idx_prod_category` ON (`category_id`), `idx_prod_status` ON (`status`), `idx_prod_sku` ON (`sku`).

Table: `product_images`

Product photography and media assets.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Image ID.
`product_id`	VARCHAR(36)	FK -> `products(id)` ON DELETE CASCADE	None	Target product.
`image_url`	VARCHAR(255)	NOT NULL	None	Cloudflare R2 / S3 image URL.
`alt_text`	VARCHAR(150)	NULLABLE	NULL	Accessibility alt text.
`display_order`	INT UNSIGNED	NOT NULL	0	Carousel display order.
`is_primary`	BOOLEAN	NOT NULL	FALSE	Primary thumbnail flag.

2.4 Shopping Cart & Customer Orders Module

Table: `orders`

Customer purchase orders with financial totals and delivery states.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique order ID.

Column Name	Data Type	Constraints	Default Value	Description
`order_number`	VARCHAR(30)	UNIQUE, NOT NULL	None	Human code (e.g. `ORD-2026-10492`).
`customer_id`	VARCHAR(36)	FK -> `customers(id)` ON DELETE RESTRICT	None	Ordering customer.
`subtotal`	DECIMAL(10,2)	NOT NULL	None	Sum of item totals (Rs.).
`discount_amount`	DECIMAL(10,2)	NOT NULL	0.00	Coupon / loyalty discount.
`delivery_fee`	DECIMAL(10,2)	NOT NULL	0.00	Delivery charge (Rs.).
`tax_amount`	DECIMAL(10,2)	NOT NULL	0.00	GST amount (Rs.).
`total_amount`	DECIMAL(10,2)	NOT NULL	None	Final payable total (`subtotal + fee + tax - discount`).
`payment_status`	ENUM('UNPAID', 'PAID', 'FAILED', 'REFUNDED')	NOT NULL	'UNPAID'	Payment status.
`order_status`	ENUM('PENDING', 'CONFIRMED', 'PROCESSING', 'PACKED', 'OUT_FOR_DELIVERY', 'DELIVERED', 'CANCELLED')	NOT NULL	'PENDING'	Fulfillment state.
`delivery_address_json`	JSON	NOT NULL	None	Snapshot of delivery address at checkout time.
`delivery_slot`	VARCHAR(50)	NOT NULL	'MORNING_HARVEST_DIS	Morning / Evening dispatch slot.
`notes`	TEXT	NULLABLE	NULL	Customer delivery instructions.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Order placement timestamp.

Indexes: `idx_orders_customer` ON (`customer_id`), `idx_orders_status` ON (`order_status`), `idx_orders_created` ON (`created_at`).

Table: `order_items`

Line items within an order.

Column Name	Data Type	Constraints	Description
`id`	VARCHAR(36)	PRIMARY KEY	Unique order item ID.
`order_id`	VARCHAR(36)	FK -> `orders(id)` ON DELETE CASCADE	Parent order ID.
`product_id`	VARCHAR(36)	FK -> `products(id)` ON DELETE RESTRICT	Ordered product ID.
`quantity`	DECIMAL(8,2)	NOT NULL	Quantity ordered (e.g. 2.50 kg).

Column Name	Data Type	Constraints	Description
`unit_price`	DECIMAL(10,2)	NOT NULL	Unit price at time of purchase.
`total_price`	DECIMAL(10,2)	NOT NULL	`quantity * unit_price`.

2.5 Inventory & Stock Transaction Ledger

Table: `inventory_items`

Real-time stock balance per product or raw material.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Inventory item ID.
`product_id`	VARCHAR(36)	UNIQUE, NULLABLE, FK -> `products(id)`	NULL	If finished mushroom product.
`item_name`	VARCHAR(100)	NOT NULL	None	Name of item (Product or Raw Material).
`item_category`	ENUM('FINISHED_PRODUC 'RAW_MATERIAL', 'PACKAGING', 'EQUIPMENT')	NOT NULL	'FINISHED_PRODUCT'	Category classification.
`available_stock`	DECIMAL(10,2)	NOT NULL	0.00	Available unreserved stock.
`reserved_stock`	DECIMAL(10,2)	NOT NULL	0.00	Stock locked in open checkout/orders.
`unit`	VARCHAR(20)	NOT NULL	'KG'	Measurement unit (KG, BAGS, PCS).
`reorder_threshold`	DECIMAL(10,2)	NOT NULL	5.00	Low-stock notification trigger.
`updated_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Last balance update.

Table: `inventory_transactions`

Immutable audit trail of every stock modification (Double-Entry Ledger).

Column Name	Data Type	Constraints	Description
`id`	VARCHAR(36)	PRIMARY KEY	Unique transaction record ID.
`inventory_item_id`	VARCHAR(36)	FK -> `inventory_items(id)` ON DELETE RESTRICT	Affected stock item.
`transaction_type`	ENUM('PURCHASE', 'PRODUCTION_USE', 'HARVEST_INJECTION', 'CUSTOMER_SALE', 'WASTAGE_SPOILAGE', 'RETURN', 'MANUAL_ADJUSTMENT')	NOT NULL	Type of movement.
`quantity`	DECIMAL(10,2)	NOT NULL	Positive (credit) or negative (debit) quantity.

Column Name	Data Type	Constraints	Description
`balance_after`	DECIMAL(10,2)	NOT NULL	Exact stock balance immediately following transaction.
`reference_type`	VARCHAR(50)	NULLABLE	e.g. `ORDER`, `HARVEST`, `PURCHASE_ORDER`.
`reference_id`	VARCHAR(36)	NULLABLE	ID of corresponding business document.
`reason`	VARCHAR(255)	NULLABLE	Explanation (e.g. "Contamination discard", "Fresh flush harvest").
`performed_by`	VARCHAR(36)	FK -> `users(id)`	Actor who executed or triggered transaction.
`created_at`	DATETIME(3)	NOT NULL	Exact timestamp.

Indexes: `idx_inv_tx_item` ON (`inventory_item_id`), `idx_inv_tx_created` ON (`created_at`).

2.6 Mushroom Cultivation & Growing Bed Module

Table: `growing_rooms`

Indoor climate-controlled cultivation rooms.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Room ID.
`name`	VARCHAR(50)	UNIQUE, NOT NULL	None	e.g. "Fruiting Chamber Alpha", "Incubation Room 1".
`room_type`	ENUM('INCUBATION', 'FRUITING', 'SPAWN_LAB', 'SUBSTRATE_STERILIZATION')	NOT NULL	'FRUITING'	Room function.
`capacity_beds`	INT UNSIGNED	NOT NULL	200	Maximum bed capacity.
`target_temp_c`	DECIMAL(4,1)	NOT NULL	24.0	Ideal temperature in Celsius.
`target_humidity`	DECIMAL(4,1)	NOT NULL	85.0	Target relative humidity %.
`is_active`	BOOLEAN	NOT NULL	TRUE	Room operational flag.

Table: `production_batches`

Master batch record for every cultivation cycle.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique batch ID.
`batch_number`	VARCHAR(40)	UNIQUE, NOT NULL	None	e.g. `BATCH-2026-OYS-042`.
`mushroom_type`	VARCHAR(60)	NOT NULL	'Oyster Mushroom'	Mushroom variety.
`spawn_supplier_id`	VARCHAR(36)	NULLABLE, FK -> `suppliers(id)`	NULL	Spawn source.

Column Name	Data Type	Constraints	Default Value	Description
`spawn_quantity_kg`	DECIMAL(8,2)	NOT NULL	None	Weight of spawn inoculated.
`substrate_type`	VARCHAR(80)	NOT NULL	'Paddy Straw + Wheat Bran'	Substrate composition.
`substrate_dry_weight_kg`	DECIMAL(8,2)	NOT NULL	None	Baseline for Biological Efficiency calculation.
`number_of_beds`	INT UNSIGNED	NOT NULL	None	Total cylindrical beds inoculated.
`inoculation_date`	DATE	NOT NULL	None	Date spawn was mixed with substrate.
`expected_harvest_date`	DATE	NOT NULL	None	Projected first flush date.
`status`	ENUM('PLANNED', 'PREPARATION', 'SPAWNING', 'INCUBATION', 'FRUITING', 'HARVESTING', 'COMPLETED', 'CONTAMINATED')	NOT NULL	'PLANNED'	Current biological stage.
`total_harvested_kg`	DECIMAL(8,2)	NOT NULL	0.00	Cumulative yield to date.
`biological_efficiency`	DECIMAL(5,2)	NULLABLE	NULL	Calculated BE % upon completion.
`created_by`	VARCHAR(36)	FK -> `users(id)`	None	Initiating farm manager.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Record creation timestamp.

Indexes: `idx_batches_status` ON (`status`), `idx_batches_number` ON (`batch_number`).

Table: `growing_beds`

Physical rack and bed location tracking.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique bed ID.
`batch_id`	VARCHAR(36)	FK -> `cultivation_batches(id)` ON DELETE CASCADE	None	Parent cultivation batch.
`room_id`	VARCHAR(36)	FK -> `growing_rooms(id)` ON DELETE RESTRICT	None	Location room.
`rack_identifier`	VARCHAR(20)	NOT NULL	'RACK-1'	Rack number.
`bed_identifier`	VARCHAR(20)	NOT NULL	'BED-001'	Bed identifier tag.
`spawn_weight_grams`	INT UNSIGNED	NOT NULL	250	Spawn quantity in this bed.

Column Name	Data Type	Constraints	Default Value	Description
`status`	ENUM('COLONIZING', 'PINNING', 'FRUITING', 'EXHAUSTED', 'CONTAMINATED')	NOT NULL	'COLONIZING'	Bed status.
`contamination_type`	VARCHAR(80)	NULLABLE	NULL	e.g. "Trichoderma Green Mold", "Coprinus".

2.7 Harvest & Quality Control Module

Table: `harvests`

Daily picking and yield logging.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique harvest ID.
`batch_id`	VARCHAR(36)	FK -> `grow_batches(id)` ON DELETE RESTRICT	None	Source cultivation batch.
`harvest_date`	DATE	NOT NULL	None	Date picked.
`flush_number`	INT UNSIGNED	NOT NULL	1	Flush sequence (1, 2, 3, etc.).
`total_weight_kg`	DECIMAL(8,2)	NOT NULL	None	Gross weight picked before grading.
`grade_a_kg`	DECIMAL(8,2)	NOT NULL	0.00	Premium whole fresh mushrooms.
`grade_b_kg`	DECIMAL(8,2)	NOT NULL	0.00	Minor stems/small buttons.
`grade_c_dry_kg`	DECIMAL(8,2)	NOT NULL	0.00	Diverted to dehydration/powder.
`wastage_kg`	DECIMAL(8,2)	NOT NULL	0.00	Spoiled / discarded yield.
`notes`	VARCHAR(255)	NULLABLE	NULL	Quality observation notes.
`harvested_by`	VARCHAR(36)	FK -> `users(id)`	None	Farm staff actor.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Timestamp logged.

Indexes: `idx_harvests_batch` ON (`batch_id`), `idx_harvests_date` ON (`harvest_date`).

2.8 Procurement, Suppliers & Expenses Module

Table: `suppliers`

Vendors for spawn, straw, poly bags, and equipment.

Column Name	Data Type	Constraints	Description
`id`	VARCHAR(36)	PRIMARY KEY	Unique supplier ID.
`name`	VARCHAR(120)	NOT NULL	Supplier business name.
`contact_person`	VARCHAR(100)	NULLABLE	Sales representative name.

Column Name	Data Type	Constraints	Description
`phone`	VARCHAR(20)	NOT NULL	Contact phone number.
`email`	VARCHAR(120)	NULLABLE	Billing email.
`materials_supplied`	VARCHAR(255)	NOT NULL	e.g. "Oyster Spawn, Paddy Straw, 50-micron PP Bags".
`gstin`	VARCHAR(15)	NULLABLE	GST Tax ID.
`is_active`	BOOLEAN	NOT NULL (TRUE)	Active status.

Table: `expenses`

Comprehensive farm and operating cost ledger.

Column Name	Data Type	Constraints	Default Value	Description
`id`	VARCHAR(36)	PRIMARY KEY	UUID()	Unique expense ID.
`category`	ENUM('RAW_MATERIAL', 'ELECTRICITY', 'WATER', 'FARM_LABOR', 'PACKAGING', 'RENT', 'DELIVERY_FUEL', 'MARKETING', 'EQUIPMENT', 'MAINTENANCE', 'MISC')	NOT NULL	'RAW_MATERIAL'	Expense classification.
`amount`	DECIMAL(10,2)	NOT NULL	None	Amount in Indian Rupees (Rs.).
`description`	VARCHAR(255)	NOT NULL	None	Detailed reason for expense.
`expense_date`	DATE	NOT NULL	None	Incurred date.
`payment_method`	ENUM('UPI', 'BANK_NEFT', 'CASH', 'DEBIT_CARD')	NOT NULL	'UPI'	Payment mechanism.
`receipt_url`	VARCHAR(255)	NULLABLE	NULL	Scanned invoice / receipt image.
`recorded_by`	VARCHAR(36)	FK -> `users(id)`	None	Recording founder / staff member.
`created_at`	DATETIME(3)	NOT NULL	CURRENT_TIMESTAMP(3)	Entry timestamp.

Indexes: `idx_expenses_category` ON (`category`), `idx_expenses_date` ON (`expense_date`).

2.9 System Audit Log & Business Settings

Table: `audit_logs`

Immutable system-wide change log for complete accountability.

Column Name	Data Type	Constraints	Description
`id`	BIGINT UNSIGNED	PRIMARY KEY, AUTO_INCREMENT	Monotonic log sequence.

Column Name	Data Type	Constraints	Description
`actor_id`	VARCHAR(36)	FK -> `users(id)`	User who executed the mutation.
`action`	VARCHAR(50)	NOT NULL	e.g. `PRICE_UPDATE`, `STOCK_WRITE_OFF`, `ROLE_GRANT`.
`entity_type`	VARCHAR(50)	NOT NULL	e.g. `Product`, `InventoryItem`, `Order`.
`entity_id`	VARCHAR(36)	NOT NULL	Target entity identifier.
`old_state`	JSON	NULLABLE	Snapshot before modification.
`new_state`	JSON	NULLABLE	Snapshot after modification.
`ip_address`	VARCHAR(45)	NULLABLE	Client IP address.
`created_at`	DATETIME(3)	NOT NULL	Event timestamp.

Indexes: `idx_audit_actor` ON (`actor_id`), `idx_audit_entity` ON (`entity_type`, `entity_id`), `idx_audit_created` ON (`created_at`).